

Experiment 5: Out of Boundary Paths

Aim: To find the number of paths by which a ball can move out of the boundary of a grid within a given maximum number of moves.

Algorithm

1. Read the number of rows, columns, maximum moves, and starting position.
2. Initialize a 2D array to store the number of ways to reach each cell.
3. Start from the given position.
4. For each move, check the four directions: up, down, left, and right.
5. If the ball moves outside the grid, increment the count.
6. Otherwise, update the number of ways for the destination cell.
7. Print the total number of paths modulo 10^9+7 .

Python Program

```
def findPaths(m, n, maxMove, startRow, startColumn):
```

```
    MOD = 10**9 + 7
```

```
    dp = [[0] * n for _ in range(m)]
```

```
    dp[startRow][startColumn] = 1
```

```
    count = 0
```

```
    directions = [(1, 0), (-1, 0), (0, 1), (0, -1)]
```

```
    for move in range(maxMove):
```

```
        temp = [[0] * n for _ in range(m)]
```

```
        for i in range(m):
```

```
            for j in range(n):
```

```
                if dp[i][j] > 0:
```

```
                    for dx, dy in directions:
```

```
                        x = i + dx
```

```
                        y = j + dy
```

```
                        if x < 0 or x >= m or y < 0 or y >= n:
```

```
                            count = (count + dp[i][j]) % MOD
```

```
                        else:
```

```

        temp[x][y] = (
            temp[x][y] + dp[i][j]
        ) % MOD

    dp = temp

    return count

m = int(input("Enter number of rows: "))
n = int(input("Enter number of columns: "))
maxMove = int(input("Enter maximum moves: "))
startRow = int(input("Enter starting row: "))
startColumn = int(input("Enter starting column: "))
result = findPaths(m, n, maxMove, startRow, startColumn)
print("Number of paths:", result)

```

Input

Enter number of rows: 2

Enter number of columns: 2

Enter maximum moves: 2

Enter starting row: 0

Enter starting column: 0

Output

```

IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Sireesha/OneDrive/Documents/DAA EX-5.py =====
Enter number of rows: 2
Enter number of columns: 2
Enter maximum moves: 2
Enter starting row: 0
Enter starting column: 0
Number of paths: 6
>>>

```

Result

Thus, the Python program to calculate the number of paths that move out of the grid boundary within the given maximum moves was successfully executed.